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An Internship Report On
“DATA SCIENCE”

Submitted in Partial fulfillment for the award of degree of

Bachelor of Engineering
in
COMPUTER SCIENCE AND ENGINEERING

Submitted by
GOUTAM R DUBASHI
1MJ20CS072

Internship carried out at
“ROBOROX OMNI TECH SOLUTIONS”
Bangalore, Karnataka

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MVJ COLLEGE OF ENGINEERING
BANGALORE-67
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Internship report entitled “**DATA SCIENCE**” is a bonafide work carried out by **GOUTAM R DUBASHI(1MJ20CS072)** in partial fulfillment for the award of the degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2023 – 2024. It is certified that all the corrections/suggestion indicated for Internal Assessment have been incorporated in the report. The Internship report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I, **GOUTAM R DUBASHI**, hereby declare that the Internship titled “**DATA SCIENCE**” embodied in this report has been carried out by me during VIII Semester of B.E degree at MVJCE Bangalore affiliated to VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELGAUM. The work embodied in this report is original and it has not been submitted in part or full for any other degree in any University.

GOUTAM R DUBASHI

1MJ20CS072

Date:

Place: BANGALORE

INTERNSHIP CERTIFICATE

ROBOROSX OMNI TECH SOLUTIONS LLP

LLPIN : ABB-4783

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May 04, 2024

CERTIFICATE OF COMPLETION

Our company is pleased to award this certificate to

GOUTAM R DUBASHI

This is to certify that Goutam has successfully completed the internship program on the project titled "Fake Hotel Reviews Prediction Using Machine Learning," assigned by RoborosX Omni Tech Solutions LLP from Feb 08, 2024, to April 30, 2024.

Throughout the internship, Goutam R Dubashi took on the responsibility of checking and developing a predictive model for identifying fake hotel reviews using machine learning. Goutam's systematic approach to the classification of hotel reviews as fake or genuine, along with his thoughtful consideration of potential enhancements for future work, reflects his commitment and valuable contributions to the project.

I am pleased to acknowledge that Goutam has successfully concluded the project by developing the assigned project. His dedication and effort are appreciated.

This certificate is awarded in recognition of the successful completion of the internship program. We commend Goutam R Dubashi for his dedication, hard work, and the successful execution of the assigned project. We wish him continued success in his future endeavors.



Anoop Narayan

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The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible, success is the epitome of hard work and perseverance, but steadfast of all is encouraging guidance.

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Thanking you,

ABSTRACT

Online reviews can be deceptive or manipulative evaluations of services and products which are often carried out deliberately for manipulation strategy to mislead the readers. Identifying such reviews is an important but challenging problem. There are even some associations in the merchandise industry who are hiring professionals to write fake reviews so that they can promote their products or defame rivals products. Hence we aim to develop a method which will detect fake reviews and remove them. The proposed method classifies users' reviews into suspicious, fake, positive and negative categories by phase-wise processing. In this paper, we are processing hotel reviews by using different data mining techniques. Moreover the reviews obtained from users are being classified into positive or negative which can be used by a consumer to select a product. Organizations providing services can monitor customer sentiments by scrutinizing and understanding what the customers are thinking about products through reviews. This can help buyers to purchase valuable products and spend their money on quality products. Also in our model end users see star ratings based on reviews for each hotel.

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CHAPTER -1

INTRODUCTION

In today's data-rich world, the role of data science has become increasingly pivotal across various industries and domains. Data science is the interdisciplinary field that utilizes scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data. With the advent of big data and advancements in technology, organizations are recognizing the immense value that data science brings in making informed decisions, solving complex problems, and driving innovation.

The essence of data science lies in its ability to transform raw data into actionable insights, thereby empowering businesses to gain a competitive edge and optimize their operations. From predicting customer behavior and optimizing marketing strategies to enhancing healthcare outcomes and improving manufacturing processes, the applications of data science are diverse and far-reaching.

Data scientists play a crucial role in this ecosystem, acting as the architects of data-driven solutions. They possess a blend of skills encompassing statistics, programming, machine learning, and domain expertise, enabling them to extract meaningful patterns and trends from data. By employing techniques such as data mining, statistical analysis, and predictive modeling, data scientists unravel valuable insights hidden within vast datasets, thereby guiding strategic decision-making and fostering innovation..

Key Components of Data science:

1. Statistics and Mathematics:

Statistical analysis forms the foundation of data science, allowing practitioners to make inferences and draw conclusions from data.

Concepts such as probability theory, hypothesis testing, regression analysis, and sampling methods are essential for understanding and analyzing data distributions, patterns, and relationships.

Mathematical techniques including linear algebra, calculus, and optimization algorithms are used for data modeling, dimensionality reduction, and machine learning.

2. Programming and Software Engineering:

Proficiency in programming languages such as Python, R, and SQL is essential for data manipulation,

analysis, and visualization.

Software engineering principles and practices enable data scientists to develop robust and scalable solutions, including the design and implementation of data pipelines, APIs, and machine learning models.

Familiarity with libraries and frameworks such as pandas, NumPy, scikit-learn, TensorFlow, and PyTorch facilitates efficient data processing, modeling, and deployment.

3. Machine Learning and Artificial Intelligence:

Machine learning algorithms form the core of data science, enabling predictive modeling, pattern recognition, and automated decision-making.

Supervised learning techniques involve training models on labeled data to make predictions or classify new instances.

Unsupervised learning methods uncover hidden patterns and structures within unlabeled data, including clustering, dimensionality reduction, and anomaly detection.

Deep learning, a subset of machine learning, leverages neural networks with multiple layers to extract hierarchical representations from complex data, such as images, text, and time-series.

4. Domain Expertise and Business Acumen:

Understanding the domain-specific context and business objectives is crucial for framing data science problems, defining relevant metrics, and interpreting results in meaningful ways.

Effective communication skills enable data scientists to collaborate with stakeholders, translate technical findings into actionable insights, and drive data-driven decision-making.

Domain expertise allows data scientists to tailor solutions to specific industry challenges, whether in finance, healthcare, marketing, retail, or other sectors.

Skills Required for Data science:

1. Proficiency in programming languages such as Python, R, and SQL is essential for data manipulation, analysis, and visualization.
2. Knowledge of libraries and frameworks like pandas, NumPy, scikit-learn, TensorFlow, and PyTorch facilitates efficient data processing, modeling, and deployment.
3. Understanding of statistical concepts such as probability theory, hypothesis testing, regression analysis, and sampling methods is crucial for analyzing data distributions, patterns, and relationships.
4. Familiarity with mathematical techniques including linear algebra, calculus, and optimization algorithms is valuable for data modeling, dimensionality reduction, and machine learning.

5. Knowledge of machine learning algorithms and techniques enables predictive modeling, pattern recognition, and automated decision-making.
6. Understanding of supervised and unsupervised learning methods, as well as deep learning techniques, allows for the exploration of complex data structures and the development of advanced models.
7. Skills in data cleaning, preprocessing, and transformation are essential for preparing raw data for analysis and modeling.
8. Proficiency in data visualization tools and techniques such as Matplotlib, Seaborn, and Tableau enables the creation of informative and visually appealing visualizations to communicate insights effectively.

Career Opportunities in Data science:

Data Science offers a wide range of career opportunities including:

1. Data Scientist
2. Data analyst
3. Machine learning engineer
4. Data Engineer
5. Business Intelligence Analyst
6. AI Research Scientist

In essence facilitating the seamless execution of technical initiatives, projects, and endeavors. Whether overseeing software development projects, coordinating technical resources, or ensuring alignment between technical and non-technical stakeholders, technical coordinators play a crucial role.

CHAPTER-2

ABOUT THE ORGANIZATION

2.1 Introduction:

RoborosX : a bangalore based startup which focuses on providing web based & android based solutions in software domain, embedded system & iot solutions in mechatronics domain

RoborosX was founded by Team RosX in 2019. Back then We were a team of Engineers from Various Streams, started with 4 members Shravan, Yashwanth, Pavankumar, Anoop now we grew up to a family of 11 from different parts of INDIA in just a span of 3 years.

- **Founding and Headquarters:** Founded in 2019 and headquartered in Bangalore, India.
- **Services Offered:** Software solution: Website, App, IoT Workshops and UAV Training. At RoborosX, we believe there is a way where customers are earned rather than bought. We're obsessively passionate about it, and our mission is to help you achieve it.

2.2 Vision:

The vision of ROBOROX OMNI TECH SOLUTIONS as a technical coordination entity is to become a leading catalyst for social change by harnessing technology to empower underprivileged communities and foster sustainable development. Here's a detailed vision statement:

"At ROBOROX OMNI TECH SOLUTIONS, our vision is to harness the power of technology to create a world where every individual, regardless of background or circumstance, has equal access to opportunities and resources for a better life. We envision a society where innovation and compassion intersect to address pressing social challenges and uplift the most marginalized communities.

Our commitment to technical coordination drives us to pioneer innovative solutions that bridge the gap between technology and social impact. We strive to be at the forefront of leveraging digital tools, data analytics, and collaborative platforms to streamline operations, maximize efficiency, and amplify the reach and effectiveness of our programs.

We envision a future where our technical coordination efforts enable seamless communication, collaboration, and knowledge sharing among stakeholders, including communities, volunteers, partner organizations, and donors. By facilitating transparent and accountable processes, we aim to build trust, foster partnerships, and mobilize collective action for sustainable change.

At the heart of our vision is a deep-seated belief in the power of community-driven development. We aspire to cultivate a culture of inclusivity, empathy, and empowerment, where every individual is valued, respected, and given the opportunity to thrive. Through our technical coordination initiatives, we seek to empower communities to take ownership of their futures, drive local solutions, and create lasting impact.

As we pursue our vision, we remain guided by principles of integrity, transparency, and ethical leadership. We are committed to upholding the highest standards of accountability and governance, ensuring that our actions reflect our values and contribute positively to the well-being of society.

In summary, our vision for ROBOROX OMNI TECH SOLUTIONS as a technical coordination entity is to be a catalyst for transformative change, leveraging technology to build a more equitable, inclusive, and sustainable world for all."

2.3 Mission

The mission of ROBOROX OMNI TECH SOLUTIONS as a technical coordination entity is to empower underprivileged communities through innovative technical solutions, collaborative partnerships, and capacity-building initiatives. Here's our mission statement: "At ROBOROX OMNI TECH SOLUTIONS, our mission is to harness the transformative power of technology to address social inequities and improve the quality of life for marginalized communities. We are dedicated to advancing sustainable development goals through strategic technical coordination, community engagement, and capacity-building efforts.

2.4 Services

ROBOROX OMNI TECH SOLUTIONS, as a technical coordination entity, offers a range of services aimed at empowering underprivileged communities and driving sustainable development:

1. **Technical Coordination:** The foundation provides technical coordination services to streamline operations, enhance communication, and optimize resource allocation across various projects and initiatives. This includes project management, data analytics, and digital infrastructure support.
2. **Community Outreach:** ROBOROX OMNI TECH SOLUTIONS conducts community outreach programs to engage with marginalized communities, understand their needs, and co-create solutions tailored to their specific challenges. This involves organizing community events, conducting surveys, and facilitating participatory decision-making processes.
3. **Capacity Building:** The foundation offers capacity-building programs to empower individuals and organizations with the knowledge, skills, and resources needed to drive positive change. This includes technical training workshops, leadership development programs, and mentorship initiatives.
4. **Project Management:** ROBOROX OMNI TECH SOLUTIONS provides project management services to oversee the planning, implementation, and evaluation of social development projects. This involves setting goals, defining strategies, mobilizing resources, and monitoring progress to ensure successful outcomes.
5. **Collaboration and Partnership:** The foundation facilitates collaboration and partnership between various stakeholders, including government agencies, non-profit organizations, corporate partners, and community members. This includes forging strategic alliances, fostering networks, and coordinating joint initiatives for maximum impact.
6. **Technology Solutions:** ROBOROX OMNI TECH SOLUTIONS develops and deploys technology solutions to address specific social challenges, such as access to education, healthcare, livelihood opportunities, and environmental sustainability. This includes mobile applications, digital platforms, and online tools designed to empower communities and improve service delivery.
7. **Advocacy and Policy Influence:** The foundation advocates for policy reforms and systemic changes to address root causes of social inequities and promote inclusive development. This involves conducting research, generating evidence-based recommendations, and engaging with policymakers and other stakeholders to influence decision-making processes.
8. **Monitoring and Evaluation:** ROBOROX OMNI TECH SOLUTIONS conducts monitoring and evaluation activities to assess the impact and effectiveness of its programs and initiatives. This includes collecting data, analysing results, and using feedback to inform decision-making and continuous improvement efforts.
9. **Knowledge Sharing and Learning:** The foundation promotes knowledge sharing and learning

among stakeholders through workshops, seminars, publications, and online platforms. This includes sharing best practices, lessons learned, and success stories to inspire and inform others working in the field of social development.

10. **Sustainability and Resilience:** ROBOROX OMNI TECH SOLUTIONS integrates principles of sustainability and resilience into its programs and initiatives to ensure long-term impact and adaptability to changing conditions. This includes promoting eco-friendly practices, building local capacity, and fostering community resilience to external shocks and stresses.



Fig.-2.1: robox logo

CHAPTER -3

ABOUT THE DEPARTMENT

3.1 Introduction

The Data Science Department at ROBOROX OMNI TECH SOLUTIONS is a dynamic and innovative team responsible for orchestrating technical solutions, managing projects, and facilitating collaborations to drive positive social impact. Comprising skilled technical coordinators, project managers, community outreach specialists, and capacity-building experts, this department is dedicated to leveraging technology for the betterment of underprivileged communities.

Key Responsibilities:

1. Data Collection and Cleaning:

Gathering data from various sources such as databases, APIs, and streaming platforms.

Cleaning and preprocessing the data to remove inconsistencies, errors, and missing values.

2. Exploratory Data Analysis (EDA):

Conducting exploratory data analysis to understand the characteristics and patterns in the data.

Visualizing data using charts, graphs, and statistical summaries to identify trends and relationships.

3. Feature Engineering:

Selecting and engineering relevant features from the data to improve model performance.

Transforming raw data into meaningful features that capture relevant information for predictive modeling.

4. Model Development and Evaluation:

Developing machine learning models and algorithms to solve specific business problems.

Evaluating model performance using appropriate metrics and techniques such as cross-validation and hyperparameter tuning.

5. Model Deployment and Integration:

Deploying machine learning models into production environments for real-time prediction or decision-

making.

Integrating models with existing systems, applications, and workflows to automate processes and workflows.

6. Continuous Monitoring and Maintenance:

Monitoring model performance over time and identifying drift or degradation in model accuracy.

Retraining models periodically with new data to ensure they remain relevant and effective.

7. Collaboration and Communication:

Collaborating with cross-functional teams including data engineers, business analysts, and stakeholders.

Communicating technical findings, insights, and recommendations to non-technical stakeholders in a clear and understandable manner.

8. Ethical and Legal Considerations:

Ensuring compliance with ethical guidelines and legal regulations related to data privacy, security, and confidentiality.

Upholding ethical standards in data collection, analysis, and decision-making to mitigate biases and risks.

9. Continuous Learning and Professional Development:

Staying abreast of advancements in data science techniques, tools, and technologies.

Engaging in continuous learning, training, and professional development to enhance skills and expertise in the field.

CHAPTER -4

INTERNSHIP DOMAIN

4.1 Introduction

The Data Science Internship at ROBOROX OMNI TECH SOLUTIONS offers an opportunity for aspiring Data Science to gain hands-on experience in leveraging technology for social impact and community development. Interns will work closely with experienced Data scientist and project managers to contribute to meaningful initiatives aimed at empowering underprivileged communities and driving sustainable change.

- 1. Technical Solutions Development:** Interns will assist in designing, developing, and implementing custom technical solutions tailored to address specific challenges faced by marginalized communities. This may include web applications, mobile apps, digital platforms, and other technology-driven initiatives aimed at promoting education, healthcare, livelihood, and environmental sustainability.
- 2. Project Support:** Interns will support project managers and technical coordinators in planning, executing, and evaluating projects aimed at empowering communities and fostering inclusive development. This may involve tasks such as conducting research, gathering requirements, preparing documentation, and coordinating stakeholder engagement activities.
- 3. Community Engagement:** Interns will have the opportunity to engage directly with community members to understand their needs, aspirations, and priorities. Through participatory approaches, interns will facilitate dialogue, build relationships, and gather feedback to inform the design and implementation of community-centric solutions.
- 4. Capacity Building:** Interns will participate in capacity-building activities aimed at empowering individuals and organizations with the skills, knowledge, and resources needed to leverage technology for social good. This may include assisting in the delivery of technical training workshops, organizing digital literacy programs, and supporting mentorship initiatives.

Learning Objectives:

1. Foundational Knowledge:

Understand the fundamental concepts, principles, and theories underlying data science, including statistics, mathematics, and computer science.

2. Programming Skills:

Develop proficiency in programming languages commonly used in data science such as Python, R, and SQL.

Learn to use libraries and frameworks for data manipulation, analysis, and visualization, such as pandas, NumPy, Matplotlib, and Seaborn.

3. Statistical and Mathematical Skills:

Gain expertise in statistical techniques and methods for data analysis, hypothesis testing, and inferential statistics.

Learn mathematical concepts and algorithms used in machine learning, including linear algebra, calculus, optimization, and probability theory.

4. Machine Learning and Artificial Intelligence:

Understand the principles and algorithms of machine learning and artificial intelligence, including supervised, unsupervised, and deep learning techniques.

Gain hands-on experience in applying machine learning algorithms to real-world datasets for predictive modeling, pattern recognition, and classification tasks.

5. Data Wrangling and Preprocessing:

Learn techniques for data cleaning, preprocessing, and transformation to prepare raw data for analysis and modeling.

Explore methods for handling missing values, outliers, and inconsistencies in datasets.

6. Data Visualization and Communication:

Develop skills in creating effective data visualizations, charts, and graphs to communicate insights and findings.

Learn to use data visualization tools and techniques to present complex data in a clear, concise, and visually appealing manner.

7. Big Data Technologies:

Familiarize with big data technologies and platforms for handling and processing large-scale datasets, such as Hadoop, Spark, and Kafka.

Gain experience in distributed computing and parallel processing frameworks for analyzing massive volumes of data efficiently.

Requirements:

- Currently enrolled in a relevant academic program (e.g., computer science, information technology, social sciences, development studies, etc.).
- Strong interest in leveraging technology for social impact and community development.
- Basic proficiency in programming languages and technologies relevant to web development, mobile app development, data analysis, or digital design.
- Excellent communication skills and ability to work effectively in a team-based environment.
- Commitment to the values and mission of ROBOROX OMNI TECH SOLUTIONS, including a passion for social justice, equality, and sustainability.

Duration and Logistics:

- The internship is typically offered on a full-time or part-time basis, with flexible scheduling options to accommodate academic commitments.
- Interns may work remotely or on-site at ROBOROX OMNI TECH SOLUTIONS's headquarters in Bangalore, India depending on the internship arrangement and public health guidelines.
- The duration of the internship may vary depending on the specific project and the intern's availability, with opportunities for extension or transition to other roles within the organization based on performance and mutual interest.

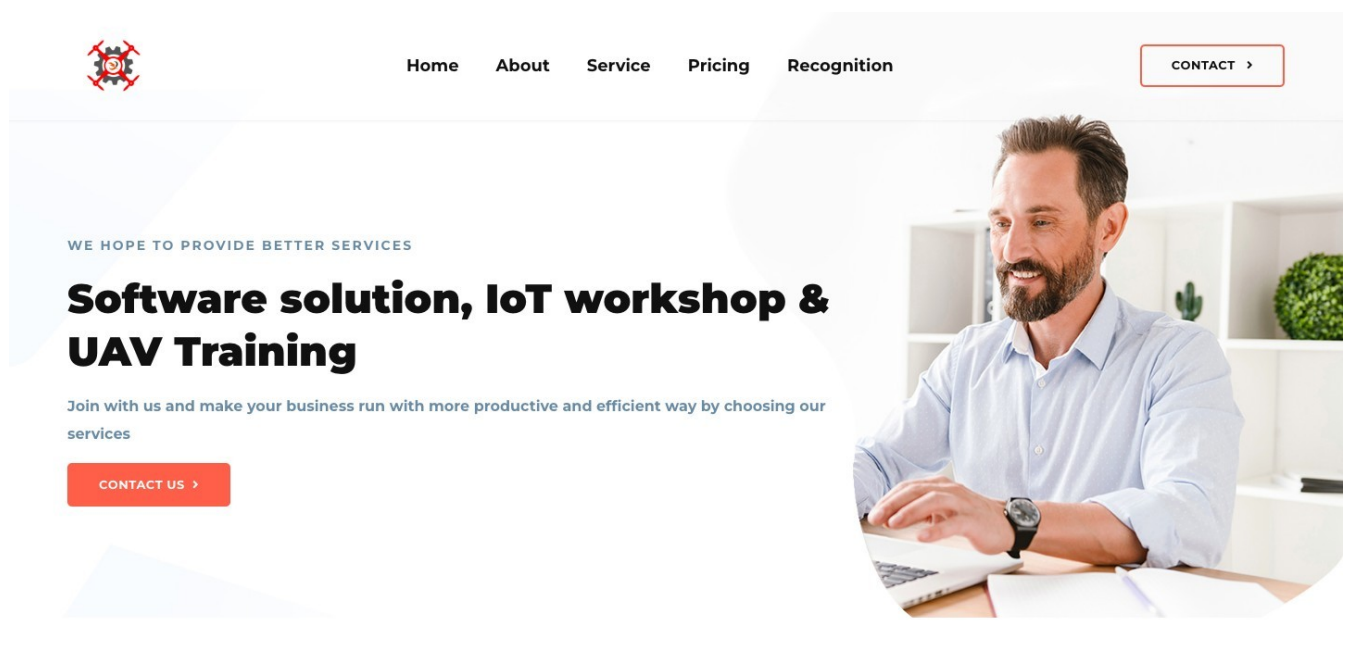


Fig.-4.1: Home Page.

**We help organizations &
enterprises and turn **their ideas into**
produtcs**



30+

Custom ERP & Android Solutions



20+

lot solutions



7+

Workshops & Internships

Fig.- 4.2: services

CERTIFICATE OF RECOGNITION



Fig.- 4.3: Certificate of Recognition

CHAPTER -7

FUTURE ASPECTS

The future of Data science is poised to be incredibly exciting, with several key trends and advancements shaping the landscape:

1. AI and Machine Learning Integration:

Continued integration of artificial intelligence (AI) and machine learning (ML) into various aspects of data science, leading to more intelligent and autonomous systems.

Advancements in deep learning techniques, reinforcement learning, and natural language processing (NLP) are expected to drive innovation in AI-driven applications.

2. Data Privacy and Ethics:

Heightened focus on data privacy, security, and ethical considerations in response to increasing concerns about data breaches, algorithmic biases, and privacy violations.

Adoption of robust data governance frameworks, transparency measures, and ethical guidelines to ensure responsible and ethical use of data.

3. Interdisciplinary Collaboration:

Growing emphasis on interdisciplinary collaboration between data scientists, domain experts, and stakeholders to address complex challenges and unlock new opportunities.

Integration of domain-specific knowledge, business expertise, and technical skills to develop holistic solutions and drive innovation in diverse domains.

4. Explainable AI and Interpretability:

Increased demand for explainable AI (XAI) and interpretable machine learning models to enhance transparency, accountability, and trust in AI-driven decision-making.

Development of techniques and methodologies for explaining model predictions, understanding model behavior, and identifying factors contributing to model outputs.

5. Edge Computing and IoT:

Proliferation of edge computing and Internet of Things (IoT) devices, leading to massive volumes of data generated at the edge of networks.

Integration of data science techniques with edge computing infrastructure to enable real-time processing, analysis, and decision-making at the network edge.

6. Augmented Analytics and Automation:

Rise of augmented analytics platforms and automated machine learning (AutoML) tools to democratize data science and empower non-technical users to analyze data and derive insights.

Automation of repetitive tasks such as feature engineering, model selection, and hyperparameter tuning to accelerate the data science workflow and improve productivity.

7. Quantum Computing and Advanced Analytics:

Exploration of quantum computing technologies to address computationally intensive problems in data science, such as optimization, cryptography, and simulation.

Potential applications of quantum computing in solving complex optimization problems, accelerating machine learning algorithms, and unlocking new possibilities in advanced analytics.

8. Data-Centric Culture and Strategy:

Shift towards a data-centric culture and strategy within organizations, where data-driven decision-making is embedded into organizational processes, workflows, and strategies.

Emphasis on data literacy, data-driven leadership, and data-driven decision-making at all levels of the organization to leverage data as a strategic asset.

These future aspects highlight the ongoing evolution of data science and its growing impact on technology, society, and business. By staying abreast of these trends and developments, data scientists can navigate the changing landscape and harness the full potential of data science to drive innovation and create value.

CONCLUSION

Working as a data scientist at Roborox Omni Tech Solutions presents an exciting opportunity to be at the forefront of technological innovation and problem-solving. Through my exploration of the company's data science department, it's evident that the role offers a dynamic and challenging environment where creativity, analytical prowess, and technical expertise are highly valued.

At Roborox Omni Tech Solutions, data scientists have the chance to collaborate with a diverse team of professionals, including engineers, researchers, and domain experts, to tackle complex challenges and drive impactful solutions. Whether it's developing machine learning algorithms for predictive maintenance in robotics or leveraging data analytics to optimize supply chain operations, the work is both intellectually stimulating and socially impactful.

Moreover, the company's commitment to fostering a culture of continuous learning and professional development ensures that data scientists have the resources and support needed to stay abreast of the latest advancements in data science and related fields. Whether through training programs, mentorship opportunities, or participation in industry conferences and events, Roborox Omni Tech Solutions empowers. Furthermore, the company's emphasis on ethical and responsible use of data underscores its commitment to integrity, transparency, and social responsibility. By upholding high ethical standards in data collection, analysis, and decision-making, data scientists at Roborox Omni Tech Solutions contribute to building trust with stakeholders and making a positive impact on society.

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